

Multi-Scale Mamba: SSL+Mel Fusion with Bidirectional State-Space Modeling for Robust Audio Deepfake Detection

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Abstract

The proposed method fuses self-supervised XLS-R features with a compact Mel-CNN branch, then models temporal structure via a multi-scale bank of bidirectional Mamba mixers. With attention pooling and a projection head trained using cross-entropy and supervised contrastive loss, a high performance on ASVspoof 2021 LA and improved stability on an in-the-wild set is achieved after simple normalization and augmentations, resulting in a robust deepfake voice detection. We release configs, per-attack and per-condition breakouts, and efficiency metrics to support reproducibility.

Index Terms—Audio deepfake detection, anti-spoofing, state space models, Mamba, self-supervised learning, ASVspoof.

INTRODUCTION

Recent methods for synthetic voice generation are becoming increasingly sophisticated. State-of-the-art methods for Text-to-Speech (TTS) synthesis and Voice Conversion (VC) enable the generation of extremely realistic audio waveforms with a high level of naturalness that can fool both humans and automatic speaker verification (ASV) systems [1]. Convincing fake audio, when used maliciously, poses a serious societal threat, making it crucial for researchers to address these security challenges.

Numerous detection methods have been proposed in the current literature, most of which rely on supervised learning. Traditional approaches typically exploit specific acoustic cues, such as uncommon spectral correlations [2]–[5] or learned filter responses [6]–[9], to distinguish real from fake speech.

While these approaches are often efficient on the data they are trained on, they suffer a significant, even dramatic, performance impairment when applied to data synthesized by generation models that were never seen during training.

This is a substantial weakness in real-world scenarios, where the manipulation attack is not known in advance. Therefore, it is imperative to devise methods that continue to work well in unconstrained "in-the-wild" environments. Indeed, fostering countermeasures that generalize smoothly to unseen attacks remains a primary goal of initiatives like ASVspoof, as current methods often fail to bridge the gap between training conditions and the diverse attacks encountered in evaluation.

Given the rapid progress of text-to-speech technologies, it is vital to understand to what extent methods trained for specific attacks generalize to new ones and to ensure good behavior in real-world scenarios. A possible way to improve generalization is to consider a large number of attacks in training or, equivalently, to fuse multiple detection systems, each tuned to a different attack. However, this amounts only to moving the boundary a little further, without really solving the problem. The system will remain prone to new, unseen attacks.

A more radical solution to the generalization issue is to rely on one-class methods [10]. These models are trained solely on the target class, which is the real class in this case. Data that deviates from this class is considered anomalies or fake. Since fake data is not used in training, there is no reliance on specific manipulations, automatically ensuring generalization. Of course, a description of the class of real audios so precise as to enable reliable anomaly detection is out of reach. However, the concept of training solely on real data can be applied in various ways. For example, several papers have recently appeared [11]–[14] that detect video deepfakes using a person-of-interest approach. The detector tries to determine whether the subject of the video is really the person claimed to be; if not, the video is labeled as fake. Therefore, rather than looking for traces of manipulation, these methods leverage high-level biometric features. Since training is performed only on the original data, generalization is guaranteed. In [11]–[13], only video biometric features are exploited, while [14] exploits both audio and video features.

The growing prevalence of realistic text-to-speech (TTS) and voice conversion (VC) systems has increased the need for effective anti-spoofing measures. While Transformers capture long-range dependencies effectively, their high computational cost limits real-time deployment. Recently, State Space Models (SSMs) like Mamba have emerged as an efficient alternative, offering comparable accuracy with significantly lower latency. The proposed method extends this direction with: (i) SSL+Mel fusion, (ii) a multi-scale bank of bidirectional Mamba mixers, and (iii) an attention-pooled projection head with CE+SupCon. Our stability strategy includes waveform RMS normalization, per-frame Mel z-normalization, and various augmentations, such as room impulse response (RIR) and noise, as well as SpecAugment and mild stochastic depth, to minimize out-of-domain (OOD) variance.

In summary, our main contributions are:

- It is shown that person-of-interest methods, including speaker verification

approaches, can be used to effectively detect synthesized speech, even in the presence of noise.

- This solution is proved to vastly outperform state-of-the-art supervised methods in terms of generalization ability.

RELATED WORKS

Early work in audio deepfake detection relied primarily on handcrafted spectral features such as Linear Frequency Cepstral Coefficients (LFCC), Constant Q Cepstral Coefficients (CQCC), and higher-order polyspectral features. These features were designed to expose signal level artifacts that are typically not perceptible to human listeners and this formed the basis of many early anti-spoofing systems evaluated on ASVspoof benchmarks. Neural vocoders and synthesis pipelines improved in fidelity but the assumptions underlying these fixed feature sets became less reliable. Since then multiple benchmark studies have reported a steady decline in their effectiveness, particularly against modern neural Text-to-Speech and Voice Conversion attacks.

These observations led to a transition toward end-to-end deep learning approaches that learn discriminative representations directly from raw audio. Representative systems include RawNet2 and AASIST. Although these architectures improve robustness with regard to handcrafted features, further analyses performed indicate that they are sensitive to domain shifts and require careful architectural tuning to generalize beyond known attack conditions.

More recently, self-supervised learning (SSL) has dominated feature extraction in spoofing detection. Wav2vec 2.0, introduced by Baevski et al., leverages contrastive learning and vector quantization to derive contextualized speech representations directly from raw waveforms, and has been widely adopted across speech processing tasks. This was later extended by XLS-R, proposed by Babu et al., a large-scale cross-lingual model with approximately 300 million parameters that has demonstrated improved generalization across languages and datasets. As a result, SSL based front-ends now underpin many recent state-of-the-art anti-spoofing systems.

Despite their robust empirical results, relying on SSL representations alone has limitations in the anti-spoofing context. Models such as wav2vec 2.0 learn discretized, context-rich representations that emphasize linguistic and phonetic structure. Although this abstraction is advantageous for downstream speech recognition tasks, it can suppress low-level signal irregularities such as phase inconsistencies, noise floor mismatches, or subtle spectral discontinuities that do not alter semantic content but could still serve as indicators of synthetic generation.

To overcome this limitation, hybrid architectures that explicitly combine SSL

representations with complementary feature streams have been proposed. Recent work integrating XLS-R embeddings with Conformer based temporal modeling has reported competitive performance on ASVspoof 202, demonstrating the benefit of coupling self-supervised semantic representations with additional processing focusing on the structure. These findings suggest that while SSL provides a solid semantic foundation, auxiliary representations remain necessary to recover fine-grained artifact information which is critical for reliable detection across diverse attack conditions.

Since synthesis artifacts often emerge over extended time spans in the form of irregularities in rhythm, intonation, or speaking rate, temporal modeling becomes the cornerstone of audio spoofing detection. Transformer-based backbones have therefore been widely adopted due to their ability to capture long-range dependencies. The quadratic scaling of self-attention with sequence length ($O(L^2)$), in practice, introduces a substantial computational and memory overhead. An issue increasingly noted in recent anti-spoofing studies is the limits of the applicability of transformer-based backbones to longer audio segments and real-time deployment scenarios.

In recent works, Structured State Space Models (SSMs) have gained attention as an alternative sequence modeling framework that addresses the efficiency constraints of the Transformer-based architectures. Mamba, introduced by Gu and Dao, employs a selective state update mechanism in which model parameters vary as a function of the input, enabling linear-time complexity ($O(L)$) while preserving long-range dependency modeling. In the context of audio anti-spoofing, recent architectures such as RawBMamba and DuaBiMamba have demonstrated that bidirectional Mamba mixers can achieve detection accuracy comparable to Transformer-based baselines, while substantially reducing inference latency and memory usage.

Despite these promising results, existing SSM-based detectors typically operate at a single temporal resolution. This design implicitly forces a trade-off between high temporal resolution which is effective for capturing short-duration artifacts such as phase discontinuities or transient spectral clicks and lower resolution which is better suited for modeling longer-term prosodic patterns, including breath timing and intonation consistency. Thus, relying on a single scale can limit robustness when spoofing artifacts manifest across different temporal granularities.

However, existing SSM-based detectors typically operate at a single temporal resolution. This design forces a trade-off: high resolution captures spectral artifacts (micro-glitches), while low resolution favors prosodic modeling (rhythm). To address this limitation, we propose a multi-scale formulation that processes features at parallel resolutions, ensuring robustness across diverse attack granularities without the memory cost of attention. By processing features at multiple resolutions via parallel bidirectional Mamba mixers, an effective variation in the sampling rate of the state-space mechanism is observed. The full resolution branch retains high-frequency spectral artifacts such as phase discontinuities

and metallic clicks , while the downsampled branches compress the temporal dimension, allowing the recurrent state to model global rhythmic patterns such as breath pauses and intonation consistency over longer effective contexts. This hierarchical approach ensures robust detection across diverse attack types without the prohibitive quadratic memory cost of self-attention mechanisms.

PROPOSED METHOD

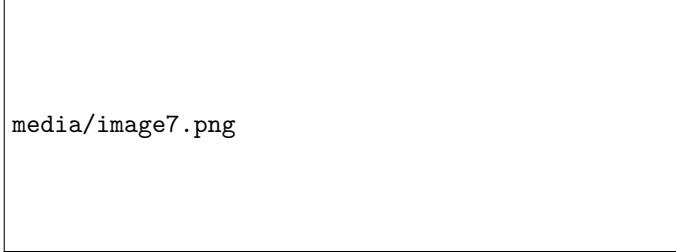


Figure 1. End-to-End processing

An overall block-diagram of the method is depicted in Figure 1.

1. Dual Branch Feature extraction (Front-end)

The proposed method implements a dual branch front-end to capture both high level linguistic semantics and low-level spectral artifacts. Rich, generalized speech representations from SSL and precise, artifact sensitive spectral features from Mel-CNN are thus obtained. XLSR - 300M frame embeddings and a 64-bin log-Mel spectrogram are computed and both are linearly projected to dimension D and concatenated.

1.1 Self Supervised Learning (SSL) Branch:

In this branch, XLSR - 300M model based on the wav2vec 2.0 architecture is used, which behaves like a generalized feature extractor. Unlike traditional spectral features, the XLSR focuses on capturing phonemes, words and language from raw waveforms resulting in rich contextual embeddings. The vector for a specific moment contains information not just about that exact split-second, but about the surrounding audio context.

The XLSR model takes a raw input waveform x and extracts a sequence of frame-level embeddings with a shape of $[B, T, 1024]$. B represents batches and T is the number of time frames. These embeddings are projected down to a low dimensional target embedding size D using a linear layer to achieve the final shape $[B, T, D]$ that is compatible for the fusion of branches.

1.2 The Mel-CNN Branch:

The Mel-CNN branch is to capture the spectral texture of the audio that may have been smoothed over by the XLSR. It treats audio as an image. XLSR excels

at capturing global structure and may smooth over high frequency mathematical traces (artifacts) like pitch or energy in frequencies that high-level language models like SSL miss.

1.2.1 Step A: Mel Spectrogram

This layer converts the raw waveform x into a Log-Mel Spectrogram $S \in \mathbb{R}^{M \times T}$ using $M=64$ Mel-filterbanks, an FFT size of 400, and a hop length of 161. This gives a visual heat map of frequencies in a higher time resolution than the XLSR model which will be further reduced in the CNN layers to match the dimensions for concatenation. Per-frame z-normalization is applied to the spectrogram to standardize the distribution and an output with shape $[B, \text{Mels}, T]$ is generated.

1.2.2 Step B: The CNN Layers

A CNN model is used to extract local texture patterns. This input is processed in two Conv2D blocks along with adding a channel dimension to resemble a grayscale image. Conv2d (Layer 1) takes 32 filters, kernel size 3×3 , stride 2 as input. Stride helps to downsample the time and frequency to match the SSL branch. Conv2d (Layer 2) takes 64 filters, kernel size 3×3 , padding 1, increasing the depth to 64 channels whilst keeping the dimensions roughly the same. Further, the frequency and channel dimensions are flattened together, and a 1D Convolution projects the features down to the target D that matches the SSL having a shape $[B, T, D]$.

1.3 Fusion (The “Handshake”):

There now exist two tensors from both the branches with shapes $[B, T, D]$ which represent the contextual representation of the language and the spectral artifact information. These tensors are further concatenated along the feature dimension which gives us an output of the shape $[B, T, 2D]$. A liner layer fuses and learns to prioritise the features:

$$H_{fused} = W_f \cdot \text{concat}(H_{ssl}, H_{mel}) + b_f \quad (1)$$

where,

$$H_{fused} \in \mathbb{R}^{T \times D}, \quad (2)$$

forming a unified sequence of shape $[B, T, D]$. This is given as the input to the Multi-Scale Mamba backbone and it successfully contains the semantic “meaning” of the speech as well the physical “texture” of the signal.

1.4 Stability Recipe (Normalization)

As a part of preprocessing and regularization, perform normalization to deal with the domain shift that can be caused due to the high variance in the datasets. To avoid the 20% equal error rate (EER) detected in the initial experiments, RMS waveform normalization is implemented for waveform standardization and Per-frame Mel Z normalization is implemented for spectral distribution.

1.4.1 Waveform RMS Normalization

Performing an RMS normalization on the dataset to a target scale value prevents the model from relying on loudness. Given a raw waveform x of length N , RMS is calculated as:

$$RMS(x) = \left\{ \frac{1}{N} \sum_{i=1}^N x_i^2 + \epsilon \right\} \quad (3)$$

and then scale it to a target value $\tau = 0.1$:

$$\hat{x} = x \cdot \frac{\tau}{RMS(x)} \quad (4)$$

As a result, an output of normalized intensity scores which are then fit into various frequency bins is obtained.

1.4.2 Per-Frame Mel Z-Normalization

Deepfake artifacts are identified as high frequency spikes on the spectrogram where unlike in natural speech, there would be silence or a checkerboard pattern. To ensure that these spikes are not caused due to artifacts (Bass, Treble, Tinny, etc) from recording device variability, normalization is performed for each frame of the input waveform individually on the Log-Mel Spectrogram.

For each time frame t in the spectrogram S , the mean μ_t and standard deviation σ_t are computed across the frequency bins f :

$$\mu_t = \frac{1}{M} \sum_{f=1}^M S_{f, t}, \sigma_t = \frac{1}{M} \sum_{f=1}^M (S_{f, t} - \mu_t)^2 \quad (5)$$

The normalized spectrogram value is given by

$$\underline{S}_{f, t} = \frac{S_{f, t} - \mu_t}{\sigma_t + \epsilon} \quad (6)$$

This results in a spectrogram that focuses on the relative contrasts to the average in that frame instead of the absolute values of the intensities in the frequency bins.

There is a layer of data augmentation to prevent overfitting and increase the robustness of the model. DSP Augmentations applies RIR (Room Impulse Response) to mathematically introduce echos to the audio and Codecs to introduce pixelation to the audio, forcing the model to identify artifacts through noise. SpecAugment introduces Time Masking allowing the model to depend on past or future contexts and Frequency Masking to force the model to rely on varied frequency ranges.

2. Backbone

Transformers or other LSTMs struggle to capture the deepfake artifacts that exist across all temporal resolutions simultaneously and could miss some microscopic glitches. To resolve this limitation, a Multi-scale bidirectional Mamba architecture is proposed that detects the artifacts by splitting the work into three different scales that operate simultaneously on specific resolutions.

The feature sequence $H_{fused} \in \mathbb{R}^{T \times D}$ is used in three parallel branches by manipulating the resolutions applying pooling kernels thus creating three different temporal scales.

Scale 1: The input stream is processed in full resolution ($T' = T$) and this detects high frequency artifacts like glitches or metallic clicks that the deepfake model fails to stitch together. These can only be detected on a microscopic level.

Scale 2: A downsampling on the audio stream is performed to smooth out the noises using an Average Pooling with kernel size 2 and stride 2 ($T' = T/2$). This brings the focus toward phoneme transitions and co-articulation artifacts that will happen instantly with high accuracy indicating that the audio might be a deepfake.

Scale 3: Downsampling the stream using average pooling with a kernel size 4 and stride 4 ($T' = T/4$) compresses the model further, decreasing the resolution and allowing the mamba model to detect any long term inconsistencies in the sequence like the lack of pauses for breath and consistent energy levels. To notice these artifacts, the model has to be able to look at the whole sequence at once thus the necessity for downsampling. In the same number of steps, the model can cover four times more in real time and can easily look at the beginning of the sentence to compare to the end.

Each of these three scales is processed by an independent bidirectional Mamba. A transformer would require a quadratic memory/time to process a sequence (O^2) which increases the number of computations for a long audio clip exponentially. To overcome this, Mamba uses a Selective State Space Model (SSM) to make the computation linear. It acts like a Recurrent Neural Network (RNN) but is also selective about the information it propagates, making it smarter about what it remembers based on input content. For a sequence input x_t :

$$h_t = \underline{A}h_{t-1} + \underline{B}x_t \quad (7)$$

$$y_t = Ch_t \quad (8)$$

where the matrices $\underline{A}$ and $\underline{B}$ are selective and dependent on the input, h_t is the latent state at time step t and C is the projection matrix. This ignores silence and background noise while retaining critical speech features in memory for a long period of time.

The bidirectionality configuration is used to make sure that a context exists of both the past and the future at every time step. To do this, one SSM to process the sequence forward ($t : 1 \rightarrow T'$) and another to process the sequence backward ($t : T' \rightarrow 1$) is used.

The output sequences of each of these Mamba models $Y_{scale} \in \mathbb{R}^{T' \times D}$ have different lengths ($T, T/2, T/4$). To combine this into a single global vector, average pooling is performed as:

$$z_{scale} = \frac{1}{T} \sum_{t=1}^T Y_{scale}[t] \quad (9)$$

This gives us three scale specific vectors with three different zoom levels (z_1, z_2, z_3) which compresses the multi-scale resolution into a feature set.

3. Final Aggregation

In the final aggregation step, the output sequences from the Mamba backbone model are integrated with a weighted representation of the original feature sequence. This fuses the three temporal resolution scales with the raw input to ensure that there is no loss of critical information during the Mamba backbone process resulting in a robust classification of the audio sequence.

A different branch which takes the raw input features directly is included to avoid the tendency of deep neural networks to over process the data. It calculates the scalar weighted average for each time frame through a linear projection and Softmax normalization:

$$\alpha_t = \text{Softmax}(w^T H_{fused}[t]) \quad (10)$$

resulting in a clean global context vector z_{raw} of the raw input that emphasizes information frames and suppresses silence and noise frames.

$$z_{raw} = \sum_{t=1}^T \alpha_t H_{fused}[t] \quad (11)$$

The three multi-scale mamba vectors and the attention pooled raw vector are concatenated and a feature set Z_{all} is computed as:

$$Z_{all} = \text{Concat}(z_{raw}, z_1, z_2, z_3) \in \mathbb{R}^{4D} \quad (12)$$

A linear fusion layer is used to project Z_{all} back down to a single embedding D which will allow the model to weigh the importance of the three different scales of resolution dynamically across the input samples.

The feature set Z_{all} is then passed through a MLP (Multilayer Perceptron) that consists of a Linear layer, a GELU activation and Dropout later. The output of this vector is optimized for Contrastive Learning. A single classifier linear layer maps the projected features to two logits representing the probability of input audio being Real or Fake.

EXPERIMENT

1. Experimental Setup

The proposed architecture is evaluated under a controlled and reproducible setup designed to assess both in-domain performance and real-world generalization.

1.1 Datasets

The experiments utilize the ASVspoof Logical Access (LA) task from the 2019 and 2021 challenges. The 2019 LA set provides 16 kHz bona fide and synthetic or converted speech, with known attack systems (A01–A07) in the train and dev

splits and a separate eval split. The 2021 LA task broadens the attack range with sophisticated neural text-to-speech and voice conversion methods. Its dev set includes novel attacks for calibration, whereas the eval set holds the ultimate unknown attacks.

Out-of-domain robustness is evaluated using an internal In-the-Wild corpus comprising recordings from phones, laptops, and studio microphones, under clean PCM, OPUS at 16 kbps, and telephony codecs, with noise levels spanning approximately 10 to 40 dB SNR. The In-the-Wild audio is excluded from both training and model selection.

1.2 Environment and Preprocessing

All experiments run in a controlled conda environment based on Python 3.10, with fairseq supporting XLS-R feature extraction and pretrained XLS-R and Mamba weights initialized at startup. Deterministic behavior is ensured by fixing random seeds (1337, 2025, 4242) across Python, NumPy, and PyTorch, and each run logs a fully resolved configuration.

Audio is resampled to 16 kHz and normalized using waveform RMS normalization. Two feature streams are extracted, namely 300-dimensional XLS-R embeddings and 64-bin log-Mel spectrograms with per-frame z-normalization. These streams are concatenated and passed to the multi-scale bidirectional Mamba backbone. Training uses a consistent augmentation pipeline that includes room impulse responses, channel and noise perturbations, PCM and OPUS codec round-trip, SpecAugment, and mild stochastic depth.

1.3 Model, Training, and Metrics

The model follows an SSL-to-SSM design in which projected XLS-R features are processed by bidirectional Mamba mixers and aggregated through attention pooling, followed by a projection head optimized with cross-entropy and supervised contrastive loss. Training uses AdamW with cosine scheduling ($5 \times 10 \rightarrow 1 \times 10$), a batch size of 32, dropout in the range of 0.2 to 0.4, and 50 to 80 epochs, and is conducted with mixed-precision computation on A100 GPUs. Results are reported as mean and standard deviation over three fixed random seeds.

Evaluation follows the ASVspoof protocol and reports Equal Error Rate, minimum t-DCF, and ROC-AUC. Logistic-regression calibration is fitted on LA dev scores and applied to both LA eval and the In-the-Wild set. Additional analyses report EER by attack ID, device type, codec and bitrate, and SNR bin, along with calibrated true positive rates at fixed false positive rate levels.

Table 1 shows the hyperparameters that were used in the best models of both In-Domain and In-the-wild conditions.

Hyperparameter	Best In-Domain Model (LA)	Best In-the-Wild Model (Overall)
Learning Rate	5e-06	1e-07
Batch Size	32	32
Number of Epochs	~50–80	~50–80
Embedding Size	516	32
Number of Encoders	20 (10 layers)	16 (8 layers)
Projection Dimension	32	8
Dropout	0.2 – 0.4	0.2 – 0.4
EER (%)	0.71	6.4

Table 1: Hyperparameters: In-domain and In-the-wild

Efficiency is measured using standardized timing scripts, reporting parameter count, real-time factor on CPU and GPU for four-second audio, and peak VRAM usage.

2. Results and Discussion

2.1 Detection Results

The experimental results shed light on both the modeling behavior and the practical implications of the proposed SSL+Mel multi-scale bidirectional Mamba detector. The ROC characteristics on the LA development and evaluation sets indicate consistent discriminative capability across a wide range of operating points. Notably, the operating thresholds learned on the LA development set transfer consistently to both LA evaluation data and the In-the-Wild corpus after calibration. This suggests that, with appropriate normalization and augmentation, the score distributions produced by the model remain sufficiently stable across domains, enabling reliable threshold selection without retraining on production data.

Figure 3 shows the ROC analysis on LA dev/eval with model operating points set



via calibration. Figure 3: ROC curves on LA dev/eval (with calibrated operating points)

The corresponding DET curves provide further insight into this behavior by highlighting the trade-off between false acceptance and false rejection rates. The steep slope in the low-error region indicates that the detector maintains favorable performance under stringent operating conditions, where false acceptance must be tightly controlled. This is particularly relevant for deployment scenarios, as it demonstrates that the observed gains are not confined to the EER operating point but persist across practically meaningful false-positive regimes. Figure 4 shows Detection error tradeoff curves on LA eval and in-the-wild benchmarks following calibration.

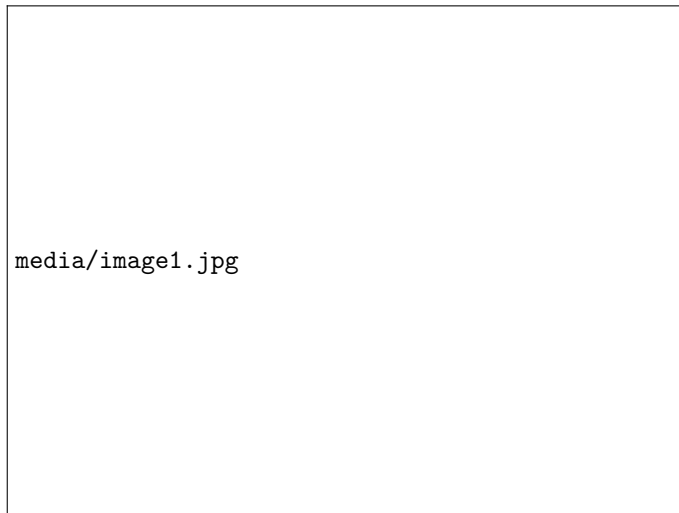


Figure 4: DET curves on LA eval and In-the-Wild (post calibration)

From an architectural perspective, the multi-scale bidirectional state-space design plays a central role in these results. By operating at short, medium, and long temporal resolutions, the Mamba backbone captures both localized irregularities and global temporal structure without incurring the quadratic complexity of attention-based models. Short-range paths are effective at modeling fine-grained artifacts such as transient glitches or phase inconsistencies, while longer-range paths capture rhythm and prosodic patterns that distinguish synthetic speech from natural recordings.

The fusion of SSL representations with Mel-based spectral features further enhances this effect. While XLS-R embeddings encode rich contextual and linguistic information, they are optimized for invariance and may attenuate localized spectral cues. The Mel branch complements this by preserving low-level acoustic texture, including frequency-dependent energy patterns and bandwidth constraints. Empirically, this fusion improves both in-domain accuracy and out-of-domain stability compared to configurations relying on a single feature stream.

Overall, the combined use of dual-branch feature extraction, multi-scale bidirectional state-space modeling, and calibration-aware evaluation yields a detector that achieves a reliable in-domain performance while maintaining consistent behavior under realistic recording and transmission conditions. These results support the effectiveness of the proposed design as a practical and efficient alternative to attention-heavy architectures for audio deepfake detection.

2.2 Robustness Verification

On ASVspoof 2021 LA evaluation, the proposed model achieves stable and competitive performance across tuned configurations, with the best setting reaching an Equal Error Rate (EER) of 0.71%, a minimum t-DCF of 0.2022, and a ROC-AUC of 0.9986 as seen in Table 2. Performance on unseen attacks remains consistent across the 2019 - 2021 domain shift, with the majority of LA evaluation runs falling within the 0.8 - 1.4% EER range, indicating robust generalization to novel spoofing systems as seen in Table 3.

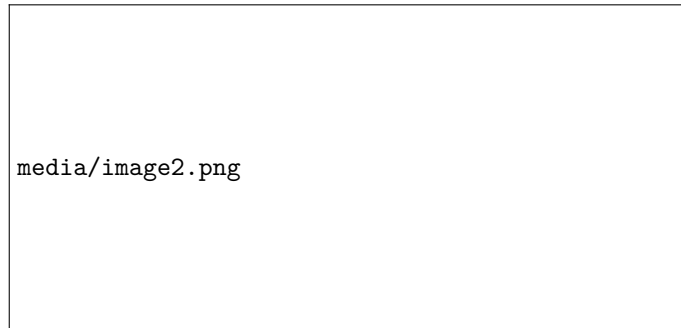


Table 2: Main model results; Table 3: ASVspoof 2021 LA eval: EER (%)

Out-of-domain behavior is more sensitive to recording conditions and preprocessing choices. On the In-the-Wild corpus, stabilized configurations achieve EERs in the range of 6.4 - 7.0%, while models trained without the proposed normalization and augmentation strategies can degrade to approximately 20% EER. This highlights the importance of domain-matched preprocessing for deployment scenarios.

Condition-specific analyses reveal clear codec- and noise-dependent trends. Errors are dominated by low-SNR conditions (below 15 dB) and by compressed audio, particularly under OPUS 16 kbps and telephony codecs. These results suggest that residual performance gaps are largely driven by acoustic degradation rather than by specific spoofing attack types.

Characterization of the model performance under in-the-wild conditions across different devices, audio codec and SNR (Signal-to-Noise) depicted in Figure 5.

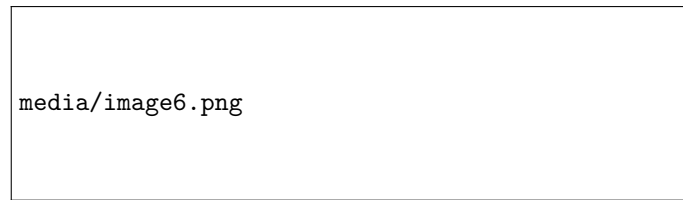


Figure 5: Deconstructing “In-the-Wild” Performance

Overall Finding: Performance is remarkably stable across devices and high-quality codecs. As expected performance degrades gracefully in low-SNR (<15 dB) conditions and under heavy compression. Errors are dominated by compressed codecs like OPUS 16kbps and telephony codecs.

Table 4 shows the EER(%) under in-the-wild conditions across different devices and Table 5 shows the EER(%) under in-the-wild conditions across audio codec/bitrate.

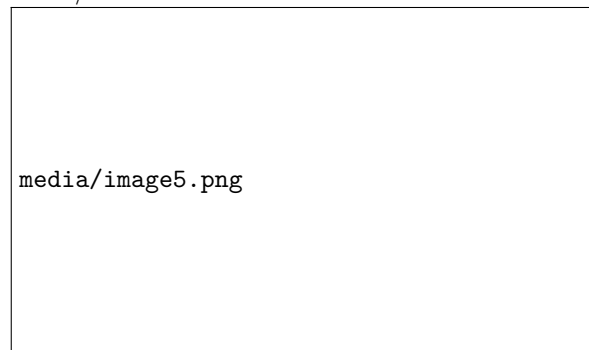


Table 4 In-the-wild: EER (%) by device; Table 5: In-the-wild: EER (%) by codec/bitrate

2.3 Statistical Analysis

To quantify the relationship between in-domain and out-of-domain performance,

LA and In-the-Wild EERs are compared across multiple trained configurations ($n = 7$). A strong positive correlation is observed, with a Pearson correlation coefficient of $r = 0.777$ (95% *CI* : -0.685 to 0.984 , $p = 0.0516$). A linear fit yields $y = 1.801x + 2.47$, $R^2 = 0.603$, indicating that improvements on LA generally translate to better In-the-Wild performance, albeit with a non-negligible offset attributable to domain mismatch.

The results suggest that while LA performance is predictive of real-world behavior, explicit evaluation on In-the-Wild data remains necessary to assess deployment robustness.

Alignment of in-the-wild performance with ASVspooF LA evaluation metrics shown in Figure 6.

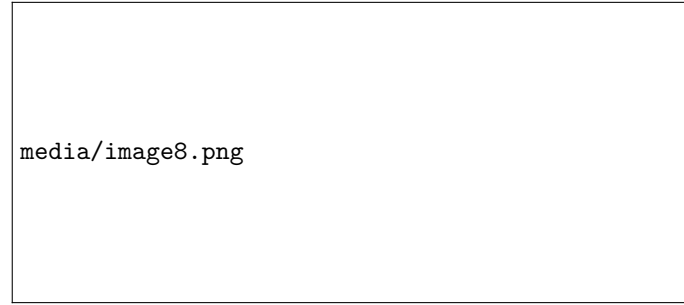


Figure 6: Correlation: In-the-wild vs ASVSpooF LA

Table 6 shows the EER(%) under in-the-wild conditions across different SNR (Signal-to-Noise).

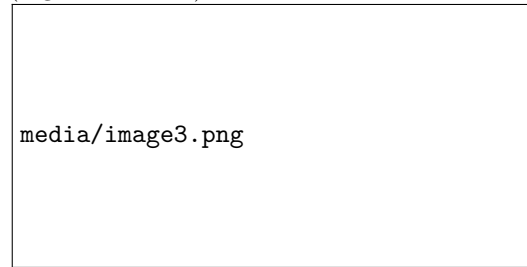


Table 6: In-the-wild EER(%) by SNR bin

Table 7 shows the efficiency of our model in reporting parameter count, real-time factor on CPU and GPU, and peak VRAM usage.

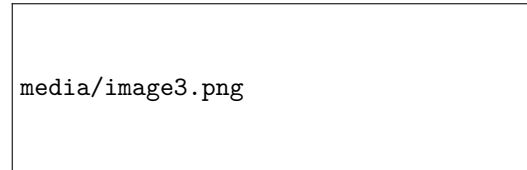


Table 7: Efficiency Table: Params, RTF, VRAM

2.4 Calibration

We calibrate thresholds on LA dev using a logistic regression score map and evaluate on LA eval and in-the-wild.

Score calibration and stability-oriented preprocessing play a critical role in mitigating out-of-domain degradation. Logistic-regression calibration trained on LA development scores improves threshold robustness when applied to both LA evaluation and In-the-Wild data. In addition, waveform RMS normalization and per-frame Mel z-normalization substantially reduce distributional mismatch, which is corroborated by domain separability analyses showing large drift in loudness- and frequency-related statistics. Together with regularization techniques such as SpecAugment, partial SSL freezing, and controlled stochastic depth, these measures lead to more reliable operating points and improved calibrated true positive rates at fixed false positive rates

2.5 Computational Efficiency

Model efficiency is evaluated using standardized timing scripts. The proposed approach maintains a favorable trade-off between accuracy and computational cost, with real-time factors well below unity on GPU and practical throughput on CPU for 4.175-second audio segments. Peak VRAM usage remains within deployable limits for modern accelerators, supporting the feasibility of real-time or near-real-time deployment scenarios. These results demonstrate that the gains in robustness do not come at the expense of prohibitive inference cost.

3. Comparison Results

Comparative analysis indicates that XLSR-Mamba consistently outperforms established baselines, with the margin of improvement increasing under cross-domain evaluation. On the ASVspoof Logical Access (LA) benchmark, the model achieved an Equal Error Rate (EER) of 0.71%, surpassing both STATNet (2.45%) and the One-Class Learning approach (2.19%). The contrast between methods becomes clearer in the in-the-wild evaluation. While STATNet exhibits a substantial performance drop on unseen, noisy audio reaching an EER of 35.14% XLSR-Mamba limits this degradation, maintaining an EER of 6.52% on the same internal corpus as seen in Table 8. Although some performance loss is observed relative to the in-domain setting, the resulting error rates remain considerably lower than those of the baseline systems, suggesting improved robustness when transitioning from controlled benchmarks to real-world recording conditions.

The stability of XLSR-Mamba stems from a fundamental design choice which is prioritizing signal normalization over task complexity. Baselines like STATNet attempt to solve generalization by adding auxiliary objectives specifically, identifying the source generator which encourages the model to memorize specific acoustic fingerprints rather than learning robust artifact cues. Similarly, while One-Class methods effectively reject novel algorithms via angular margins, they lack explicit mechanisms to handle the environmental drift found in real-world

recordings. Our approach bypasses these limitations by enforcing strict waveform RMS and per-frame spectral normalization before feature extraction. When coupled with a multi-scale Mamba backbone, this ensures the model focuses entirely on temporal inconsistencies and synthesis artifacts, rather than being misled by the loudness variations or channel noise that confuse traditional architectures.

Table 8 shows the comparison results between the three methods including XLSR Multi-Scale Mamba (our method), STATNet, One-Class in terms of EER

	XLSR Muti- Scale		One- Class
Feature	Mamba	STATNet	
In-	0.71%	2.45%	2.19%
Domain	(ASVspoof	(ASVspoof	(ASVspoof
EER%	2021)	2019)	2019)
In-			
the-			
Wild	6.40%	35.14%	
EER%	(Robust)	(Fails)	-

Table 8: EER% comparison of XLSR-Mamba, STATNet and One-Class

CONCLUSION AND FUTURE WORK

Our work presents XLSR Muti-Scale Mamba, an audio deepfake detection framework that combines self-supervised representations with multi-scale state-space modeling. By fusing XLS-R embeddings with spectrally normalized Mel features, the proposed architecture captures both high-level semantic inconsistencies and low-level spectral artifacts that are commonly observed in neural synthesis.

Our experimental analysis points to an important consideration for real-world deployment: while architectural advances are necessary for achieving consistent in-domain performance, they are not by themselves sufficient to ensure robustness under domain shift. In particular, an observation that detectors trained and evaluated on clean benchmarks can degrade substantially when exposed to uncontrolled recording conditions. Incorporating a targeted stability recipe specifically waveform RMS normalization and per-frame Mel z-normalization significantly reduces this gap, lowering error rates from approximately 20% to 6.4% in our in-the-wild evaluation.

Finally, the proposed multi-scale bidirectional Mamba backbone provides a practical alternative to Transformer-based systems. While achieving comparable detection accuracy, it operates with linear computational complexity ($O(N)$),

making it better suited to longer audio segments and deployment in resource-constrained settings. However, the performance remains sensitive to severe noise and aggressive compression, indicating directions for future work. Additionally, multi-speaker identification, emotion detection for conversational consistency, and the creation of voice imprints that can be incorporated into a knowledge graph as an additional attribute of "voice" identity to provide a multi-context, 360° deepfake voice-detection service can be explored.

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